Лабораторная работа №16

Базовая защита от атак типа «brute force»

Студент: Талебу Тенке

Группа: НФИбд 02–23

дисциплина: **Администрирование сетевых подсистем (Lab 16)**

Цель работы

- Целью данной работы является получение навыков работы с программным средством Fail2ban для обеспечения базовой защиты от атак типа «brute force».

```
  Installing       : fail2ban-server-1.1.0-6.el9.noarch                                   2/5
  Running scriptlet: fail2ban-server-1.1.0-6.el9.noarch                                   2/5
  Installing       : fail2ban-firewalld-1.1.0-6.el9.noarch                                3/5
  Installing       : fail2ban-sendmail-1.1.0-6.el9.noarch                                 4/5
  Installing       : fail2ban-1.1.0-6.el9.noarch                                          5/5
  Running scriptlet: fail2ban-selinux-1.1.0-6.el9.noarch                                  5/5
  Running scriptlet: fail2ban-1.1.0-6.el9.noarch                                          5/5
  Verifying        : fail2ban-1.1.0-6.el9.noarch                                          1/5
  Verifying        : fail2ban-firewalld-1.1.0-6.el9.noarch                                2/5
  Verifying        : fail2ban-selinux-1.1.0-6.el9.noarch                                  3/5
  Verifying        : fail2ban-sendmail-1.1.0-6.el9.noarch                                 4/5
  Verifying        : fail2ban-server-1.1.0-6.el9.noarch                                   5/5

Installed:
  fail2ban-1.1.0-6.el9.noarch                    fail2ban-firewalld-1.1.0-6.el9.noarch
  fail2ban-selinux-1.1.0-6.el9.noarch            fail2ban-sendmail-1.1.0-6.el9.noarch
  fail2ban-server-1.1.0-6.el9.noarch

Complete!
[root@server ~]# systemctl start fail2ban
[root@server ~]# systemctl enable fail2ban
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/
system/fail2ban.service.
[root@server ~]# S
```

root@server:~

root@server:~ | root@server:~

```
[vagrant@server ~]$ sudo -i
[root@server ~]# tail -f /var/log/fail2ban.log
2026-03-12 20:34:42,274 fail2ban.server         [7099]: INFO    ------------------------------
------------------
2026-03-12 20:34:42,274 fail2ban.server         [7099]: INFO    Starting Fail2ban v1.1.0
2026-03-12 20:34:42,275 fail2ban.observer       [7099]: INFO    Observer start...
2026-03-12 20:34:42,295 fail2ban.database       [7099]: INFO    Connected to fail2ban persistent
 database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-03-12 20:34:42,296 fail2ban.database       [7099]: WARNING New database created. Version '4
```

customisation.local
/etc/fail2ban/jail.d

```
[DEFAULT]
bantime = 3600

#
# SSH servers
#
[sshd]
port = ssh,2022
enabled = true
[sshd-ddos]
filter = sshd
enabled = true
[selinux-ssh]
enabled = true
```

```
 database '/var/lib/fail2ban/fail2ban.sqlite3'
2026-03-12 20:37:31,146 fail2ban.jail           [7230]: INFO    Creating new jail 'sshd'
2026-03-12 20:37:31,150 fail2ban.jail           [7230]: INFO    Jail 'sshd' uses systemd {}
2026-03-12 20:37:31,151 fail2ban.jail           [7230]: INFO    Initiated 'systemd' backend
2026-03-12 20:37:31,156 fail2ban.filter         [7230]: INFO      maxLines: 1
2026-03-12 20:37:31,180 fail2ban.filtersystemd  [7230]: INFO    [sshd] Added journal match for:
'_SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session'
2026-03-12 20:37:31,180 fail2ban.filter         [7230]: INFO      maxRetry: 5
2026-03-12 20:37:31,180 fail2ban.filter         [7230]: INFO      findtime: 600
2026-03-12 20:37:31,180 fail2ban.actions        [7230]: INFO      banTime: 3600
2026-03-12 20:37:31,180 fail2ban.filter         [7230]: INFO      encoding: UTF-8
2026-03-12 20:37:31,181 fail2ban.jail           [7230]: INFO    Creating new jail 'selinux-ssh'
2026-03-12 20:37:31,182 fail2ban.jail           [7230]: INFO    Jail 'selinux-ssh' uses poller {
}
2026-03-12 20:37:31,182 fail2ban.jail           [7230]: INFO    Initiated 'polling' backend
2026-03-12 20:37:31,183 fail2ban.datedetector   [7230]: INFO      date pattern `''`: `Epoch`
2026-03-12 20:37:31,184 fail2ban.filter         [7230]: INFO      maxRetry: 5
2026-03-12 20:37:31,184 fail2ban.filter         [7230]: INFO      findtime: 600
2026-03-12 20:37:31,184 fail2ban.actions        [7230]: INFO      banTime: 3600
2026-03-12 20:37:31,184 fail2ban.filter         [7230]: INFO      encoding: UTF-8
2026-03-12 20:37:31,224 fail2ban.filter         [7230]: INFO    Added logfile: '/var/log/audit/a
udit.log' (pos = 0, hash = 0c2be31ffde8214ab616b0e504468ebe00b577f7)
2026-03-12 20:37:31,224 fail2ban.jail           [7230]: INFO    Creating new jail 'sshd-ddos'
2026-03-12 20:37:31,226 fail2ban.jail           [7230]: INFO    Jail 'sshd-ddos' uses poller {}
2026-03-12 20:37:31,227 fail2ban.jail           [7230]: INFO    Initiated 'polling' backend
2026-03-12 20:37:31,229 fail2ban.filter         [7230]: INFO      maxLines: 1
2026-03-12 20:37:31,238 fail2ban.filter         [7230]: INFO      maxRetry: 5
2026-03-12 20:37:31,239 fail2ban.filter         [7230]: INFO      findtime: 600
2026-03-12 20:37:31,239 fail2ban.actions        [7230]: INFO      banTime: 3600
2026-03-12 20:37:31,239 fail2ban.filter         [7230]: INFO      encoding: UTF-8
2026-03-12 20:37:31,245 fail2ban.filtersystemd  [7230]: INFO    [sshd] Jail is in operation now
(process new journal entries)
2026-03-12 20:37:31,246 fail2ban.jail           [7230]: INFO    Jail 'sshd' started
2026-03-12 20:37:31,249 fail2ban.jail           [7230]: INFO    Jail 'selinux-ssh' started
2026-03-12 20:37:31,259 fail2ban.jail           [7230]: INFO    Jail 'sshd-ddos' started
```

root@server:~

root@server:~ | root@server:~ | root@server:~

```
** (gedit:7296): WARNING **: 20:39:15.499: Set document metadata failed: Setting attribute metad
ata::gedit-position not supported

(gedit:7296): dconf-WARNING **: 20:39:15.517: failed to commit changes to dconf: Failed to execu
te child process “dbus-launch” (No such file or directory)
[root@server ~]# systemctl restart fail2ban
[root@server ~]# tail -f /var/log/fail2ban.log
2026-03-12 20:39:27,417 fail2ban.filter         [7328]: WARNING [apache-botsearch] Please check
a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:warn]
 [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer library
 (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070) -
 may result in undefined or erroneous behavior
2026-03-12 20:39:27,431 fail2ban.jail           [7328]: INFO    Jail 'apache-botsearch' started
2026-03-12 20:39:27,440 fail2ban.jail           [7328]: INFO    Jail 'apache-fakegooglebot' star
ted
2026-03-12 20:39:27,448 fail2ban.filter         [7328]: WARNING [apache-modsecurity] Ignoring al
l log entries older than 600s; these are probably messages generated while fail2ban was not runn
ing.
2026-03-12 20:39:27,448 fail2ban.filter         [7328]: WARNING [apache-modsecurity] Please chec
k a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:war
n] [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer libra
ry (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070)
 - may result in undefined or erroneous behavior
2026-03-12 20:39:27,449 fail2ban.jail           [7328]: INFO    Jail 'apache-modsecurity' starte
d
2026-03-12 20:39:27,461 fail2ban.filter         [7328]: WARNING [apache-shellshock] Ignoring all
 log entries older than 600s; these are probably messages generated while fail2ban was not runni
ng.
2026-03-12 20:39:27,470 fail2ban.jail           [7328]: INFO    Jail 'apache-shellshock' started
2026-03-12 20:39:27,470 fail2ban.filter         [7328]: WARNING [apache-shellshock] Please check
 a jail for a timing issue. Line with odd timestamp: [Thu Mar 12 20:25:24.302442 2026] [ssl:warn
] [pid 1369:tid 1369] AH01882: Init: this version of mod_ssl was compiled against a newer librar
y (OpenSSL 3.5.1 1 Jul 2025 (OpenSSL 3.0.7 1 Nov 2022), version currently loaded is 0x30000070)
- may result in undefined or erroneous behavior
2026-03-12 20:39:27,508 fail2ban.jail           [7328]: INFO    Jail 'sshd-ddos' started
```

Open

*customisation.local
/etc/fail2ban/jail.d

Save

```
[DEFAULT]
bantime = 3600
ignoreip = 127.0.0.1/8
#
# SSH servers
#
[sshd]
port = ssh,2022
enabled = true
```

```
** (gedit:7438): WARNING **: 20:44:29.558: Set document metadata failed: Setting attribute metad
ata::gedit-position not supported

(gedit:7438): dconf-WARNING **: 20:44:29.584: failed to commit changes to dconf: Failed to execu
te child process “dbus-launch” (No such file or directory)
[root@server ~]# systemctl restart fail2ban
[root@server ~]# tail -f /var/log/fail2ban.log
2026-03-12 20:44:38,221 fail2ban.jail          [7469]: INFO    Jail 'apache-auth' started
2026-03-12 20:44:38,247 fail2ban.jail          [7469]: INFO    Jail 'apache-badbots' started
2026-03-12 20:44:38,251 fail2ban.jail          [7469]: INFO    Jail 'apache-noscript' started
2026-03-12 20:44:38,253 fail2ban.jail          [7469]: INFO    Jail 'apache-overflows' started
2026-03-12 20:44:38,257 fail2ban.jail          [7469]: INFO    Jail 'apache-nohome' started
2026-03-12 20:44:38,294 fail2ban.jail          [7469]: INFO    Jail 'apache-botsearch' started
2026-03-12 20:44:38,317 fail2ban.jail          [7469]: INFO    Jail 'apache-fakegooglebot' star
ted
2026-03-12 20:44:38,360 fail2ban.jail          [7469]: INFO    Jail 'apache-modsecurity' starte
d
2026-03-12 20:44:38,392 fail2ban.jail          [7469]: INFO    Jail 'apache-shellshock' started
2026-03-12 20:44:38,402 fail2ban.jail          [7469]: INFO    Jail 'sshd-ddos' started
```

Open

protect.sh
/vagrant/provision/server

Save

```bash
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install fail2ban
echo "Copy configuration files"
cp -R /vagrant/provision/server/protect/etc/* /etc
restorecon -vR /etc
echo "Start fail2ban service"
systemctl enable fail2ban
systemctl start fail2ban
```

Вывод

- В ходе выполнения лабораторной работы были получены навыки работы с программным средством Fail2ban для обеспечения базовой защиты от атак типа «brute force».

Спасибо за внимание!